

Love AI More. Trust AI Less.

DSCT lecture walkthrough

Abstract

A short report can be a reasoning instrument: state a claim, inspect evidence, and decide how much confidence is earned.

1 Claim and evidence

Claim: Increasing simultaneous requests should increase aggregate AI throughput until parallelism is saturated. The table and plot are generated evidence, not decorative

support.	Hardware	parallel limit	$N = 1$	$N = 4$	$N = 8$	$N = 32$
	Tesla T4	1	69.2 / 0.49	74.1 / 0.54	73.4 / 0.53	73.4 / 0.54
	Tesla T4	8	72.6 / 0.53	102.8 / 2.18	117.3 / 4.77	121.9 / 4.84
	GTX 1060 3 GB	1	59.6 / 0.61	84.5 / 0.86	90.2 / 0.92	94.6 / 0.98
	GTX 1060 3 GB	4	73.6 / 0.99	112.1 / 3.76	120.7 / 3.98	119.9 / 3.97

Each cell: aggregate tokens/s / overlap factor.

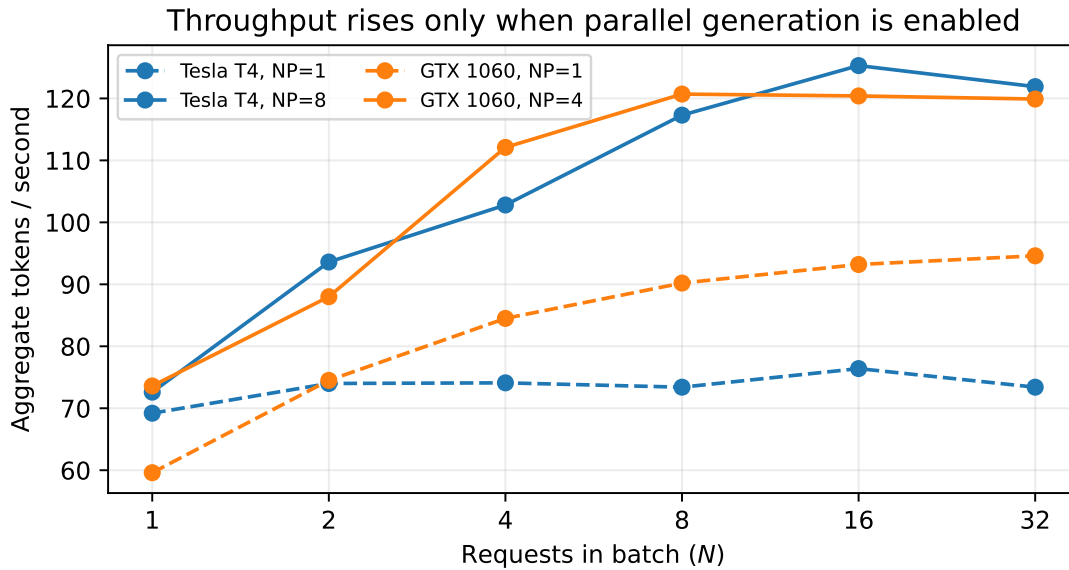


Figure 1: Aggregate throughput across six batch sizes.

2 What the evidence actually shows

With parallel limit 1, throughput approaches one stream: the GTX 1060 rises from 59.6 to 94.6 tokens/s as overhead is hidden, but it never demonstrates four requests generating at once. With limit 8, the T4 reaches

125.3 tokens/s at $N = 16$ and then levels off. With limit 4, the GTX 1060 reaches 120.7 at $N = 8$ and levels off. The knee is near the parallel limit, not automatically at the largest N .